Part D: Recommendations Report

Executive Summary

STADIOalot has been experiencing an increase in late and repeat deliveries. This report examines whether historical delivery data can help identify orders that are likely to arrive late, giving the business an opportunity to address potential problems before they affect customers.

Two models were tested using the Brazilian E-Commerce Public Dataset by Olist: Logistic Regression (Model 1) and Random Forest (Model 2). Both were evaluated using the same test set of 19,294 orders.

The results show that Logistic Regression is the better option for further development. It identified more late deliveries and performed better across most of the key measures. However, neither model is ready for immediate business use. Further improvements and testing with STADIOalot's own delivery data are needed before the model can be relied on for operational decisions.

Which model is best?

The results for the two models are summarised below.

Performance measure	Model 1: Logistic Regression	Model 2: Random Forest
Accuracy	56.41%	82.18%
Precision	9.11%	6.54%
Recall	86.07%	17.83%
F1-score	16.37%	9.57%
ROC-AUC	0.7198	0.5571

Although Random Forest achieved higher accuracy, this does not mean it was better at identifying late deliveries. Its recall was only 17.83%, meaning it missed many orders that were delivered late.

Logistic Regression achieved a recall of 86.07%, identifying a much larger proportion of late deliveries. Its ROC-AUC of 0.7198 also indicates a better ability to distinguish between orders that are late and those that are delivered on time.

However, Logistic Regression had a precision of only 9.11%. This means that many orders flagged as being at risk were not actually late. Following up on all these orders could create additional work for the delivery team.

Recommendation: STADIOalot should use Logistic Regression as the starting point for further development. It is better suited to identifying potentially late orders, but its accuracy

in flagging individual orders needs to improve. The business should also consider the cost of missing a late delivery compared with the cost of checking an order that would have arrived on time.

How can the selected model be improved?

The first improvement should focus on reducing the number of orders incorrectly flagged as late without losing too many genuine late deliveries.

The model currently works with public data that may not fully reflect the factors affecting STADIOalot's deliveries. Adding more relevant information could improve its predictions. Useful information may include the time taken to dispatch an order, delivery distance, courier performance, destination, order size, warehouse workload and previous delivery problems.

Only information available when the prediction is made should be included. For example, the actual delivery date cannot be used to predict whether an order will be late, because it is only known after delivery has taken place.

The model should also be tested using different decision thresholds. This would allow STADIOalot to decide how many orders should be flagged for follow-up, depending on the capacity of its delivery team and the cost of late deliveries.

Further testing should compare the models using several measures rather than accuracy alone. In particular, the business should monitor how many late deliveries are identified, how many flagged orders turn out to be late, and how often the model raises unnecessary alerts.

These changes should be tested on data that was not used to develop the model. This will help establish whether the improvements are likely to work in practice.

How should the model be adapted to STADIOalot's requested data?

The Olist dataset provides a useful starting point, but it cannot fully address STADIOalot's requirements. It supports the study of late deliveries, but it does not provide reliable information on individual repeat delivery attempts. STADIOalot's own data is therefore needed to investigate both problems.

The data requested in SS1 should be reviewed to confirm that it contains the information required to understand each order's journey. This should ideally include order and dispatch dates, promised delivery dates, actual delivery dates, delivery location, courier details and records of previous delivery attempts.

Once the data is received, it should be checked for missing values, duplicate records, inconsistent dates and incorrect entries. The business must also agree on what counts as a late delivery and what qualifies as a repeat delivery attempt.

The next step is to prepare the data based on the point at which STADIOalot wants to make a prediction. For example, if the goal is to identify risky orders when they are dispatched, the model should only use information available at dispatch time. This will ensure that the results reflect what the business could realistically know at that stage.

Logistic Regression can then be trained and tested using STADIOalot's historical data. The results should be compared with the current model and assessed across different couriers, locations and order types. A separate prediction target should be developed for repeat delivery attempts, since the factors contributing to a late delivery may differ from those leading to a failed delivery.

If the results are satisfactory, the business could first run a small pilot. Delivery teams could use the risk scores to prioritise follow-ups, investigate potential delays and take action where necessary. The pilot should measure whether this reduces late deliveries, repeat attempts and the time spent managing delivery exceptions.

Do the results agree with the literature reviewed in Part A?

The findings are broadly consistent with the research reviewed in Part A, which highlights the value of using historical data to support delivery planning and identify orders that may need attention.

Rokoss et al. (2024) investigated machine learning approaches to estimating delivery times. Their research highlights the importance of selecting relevant information and understanding which factors are available when making a prediction. This supports the recommendation to improve the model using reliable delivery and operational data from STADIOalot.

Madane and Azeroual (2026) examined how e-commerce orders can be assessed and prioritised for service recovery. Their work is relevant because identifying orders at risk can help a business decide where to focus its attention and resources. The results from Logistic Regression suggest that this approach could be useful at STADIOalot, provided the number of incorrect alerts is reduced.

Alves et al. (2019) explored delivery challenges and possible interventions through simulation, including failed deliveries and alternative delivery arrangements. Although their study did not focus on predicting late deliveries using the same type of classification model, it highlights the wider operational importance of addressing delivery failures.

Overall, the literature supports the direction of this study. However, the results do not prove that the model will perform equally well at STADIOalot. This can only be established through further testing with the company's own data.

References

Alves, R., da Silva Lima, R., Custódio de Sena, D., Ferreira de Pinho, A. and Holguín-Veras, J. (2019) 'Agent-Based Simulation Model for Evaluating Urban Freight Policy to E-Commerce', *Sustainability*, 11(15), article 4020. Available at: https://doi.org/10.3390/su11154020.

Madane, Y. and Azeroual, M. (2026) 'Value, Risk, and Recoverability: An Interpretable Order-Level Prioritization Framework for Service Recovery in E-Commerce', *Digital*, 6(3), article 57. Available at: https://doi.org/10.3390/digital6030057.

Olist and Sionek (2018) *Brazilian E-Commerce Public Dataset by Olist*. Kaggle. Available at: https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce.

Rokoss, A., Syberg, M., Tomidei, L., Hülsing, C., Deuse, J. and Schmidt, M. (2024) 'Case study on delivery time determination using a machine learning approach in small batch production companies', *Journal of Intelligent Manufacturing*, 35, pp. 3937–3958. Available at: https://doi.org/10.1007/s10845-023-02290-2.